# RUDRAGOUDA

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## OBJECTIVE

AI Engineer with over 2 years of hands-on experience in designing, developing, and deploying scalable AI/ML solutions across real-time perception, sensor fusion, and anomaly detection. Adept at building production-ready models using deep learning frameworks like PyTorch and TensorFlow, optimizing model performance. Proven track record of solving complex business problems with cross-functional collaboration, cutting-edge research, and strong MLOps practices. Eager to contribute to AI-driven innovation with impactful and scalable solutions.

## SKILLS

- **Programming Languages:** Python, C++
- **Machine Learning & AI:** Supervised and Unsupervised Learning, Deep Learning, Model Training, Optimization, Generative AI(GANs, Transformers), Hyperparameter Tuning.
- **Framework and Libraries:** PyTorch, TensorFlow, OpenCV, YOLO, XGBoost, PaddleOCR, Scikit-learn, GAN, ETL, ROS2
- **MLOps and Deployment:** Docker, MLflow
- **Dev Tools and Version Code:** Git, Jupyter, VS Code, Visual Studio, Google Colab
- **Cloud:** Azure, AWS
- **Soft Skills:** Problem solving, Active Listening, Attention to Detail, Time Management, Adaptability

## EXPERIENCE

**AI Engineer 2yrs 2months** — **Feb 2023 - Present**
Tardid Technologies Pvt Ltd — *Bengaluru*

**Perception Sensor Fusion**

- Developed real-time sensor fusion pipelines for Camera, LiDAR, GPS, IMU, AIS and RADAR using ROS2, ensuring precise synchronization and alignment via Kalman Filter algorithms (EKF,UKF).
- Implemented YOLO-based object detection for real-time perception, optimizing inference using OpenVINO for edge deployment.
- Designed LiDAR object detection models using PointNet and PointPillars, improving 3D object classification and localization.
- Performed Camera-LiDAR calibration using OpenCalib, enhancing sensor alignment and multi-modal data fusion.
- Improved sensor communication with ROS Bridge, reducing latency by 30% and increasing system efficiency.

**OCR and Anomaly Detection for Industrial Applications**

- Built an OCR pipeline using PaddleOCR & PyTorch, increasing text recognition accuracy by 25% in diverse conditions.
- Applied data augmentation, expanding datasets by 50%, leading to a 30% improvement in AI model performance.
- Accelerated training by 50% using NVIDIA CUDA, optimizing computer vision models for speed and efficiency.
- Developed a deep learning-based anomaly detection model, achieving 87% accuracy in detecting industrial valve malfunctions.

## PROJECTS

**Super Resolution using Generative Adversarial Network.**

- Developed a deep learning model using SR-GAN to upscale low-resolution images while preserving fine details and textures.
- Improved image quality by 50%, enhancing visual clarity for real-world applications like medical imaging and satellite image processing.
- Trained and optimized the model on Google Colab, leveraging TensorFlow & PyTorch for efficient computation.

**Technologies:** SR-GAN, Deep Learning, TensorFlow, PyTorch, Google Colab

**First-Order Motion Model for Image Animation**

- Developed a keypoint-based motion transfer model using a PatchGAN-based Conditional GAN for realistic image animation.
- Implemented a generator-discriminator framework that extracts keypoints from source and driving images to synthesize smooth motion sequences.
- Optimized perceptual loss using VGG19, multi-scale PatchGAN discrimination, and image pyramids to enhance motion realism.

**Technologies:** GANs, PatchGAN, Conditional GAN, PyTorch, VGG19, Image Pyramids, Deep Learning

**Security-Based Steganography for Data Hiding.**

- Implemented an Android application for secure text embedding in images, audio, and video using the Least Significant Bit (LSB) algorithm.
- Integrated password-protected encryption to enhance security for hidden messages.
- Deployed the app using Android Studio, with cloud storage integration via Firebase for secure data management.

**Technologies:** Android Studio, Java, LSB Algorithm, Firebase

## CERTIFICATIONS AND ACHIEVEMENTS

- DevOps, DataOps, MLOps : Digital Certificate
- Microsoft Azure Fundamentals AZ-900 Exam Prep : Digital Certificate
- Participated in the Blue Yonder hackathon in Nov 2022 and demonstrated Microservice-Based Android Application.
- 1st Place in the Technical Treasure hunt Jan 2023 in Christ University.

## EDUCATION

**R V College of Engineering Bengaluru** **2021-2023**
Master of Computer Application - 8.01 cgpa
**JSS Shri Manjunatheshwara Institute of UG and PG Studies Dharwad** **2018-2021**
Bachelor of Science (Computer Science and Electronics)